STUDYLAB HUB · COLLEGE CALCULUS I & AP CALCULUS · UNIT 2 (CHAPTER 2)

Unit 2: Differentiation Rules & Chain Rule

Power, Product & Quotient Rules · Trig/Exp/Log Derivatives · Chain Rule · Implicit & Inverse Derivatives

College Calculus I & AP Calculus — Unit 2 (Chapter 2): Differentiation Rules, Chain Rule & Implicit Differentiation

Independent, original study guide for College Calculus I & AP Calculus (informed by OpenStax open textbooks, CC BY 4.0).

1. Core Unit Roadmap

Section	Core Principle & Definitions	High-Yield Exam Application
Part 1 · Fundamental Concepts	Power, Product & Quotient Rules	Identifying core variables, boundary conditions, and conserved quantities
Part 2 · Quantitative Models & Mechanisms	Trig/Exp/Log Derivatives	Applying governing equations and predicting how system state changes under perturbation
Part 3 · Advanced Problem Solving	Implicit & Inverse Derivatives	Avoiding common sign, unit, and conceptual traps on multiple-choice and free-response questions

2. Essential Principles & Step-by-Step Analysis

1. **First-Principles Mechanism**:

- Every problem in **Unit 2 (Chapter 2): Differentiation Rules, Chain Rule & Implicit Differentiation** reduces to identifying the underlying structure, forces, rates, or invariants before plugging in numbers.
- Always state the governing principle explicitly at the start of any free-response justification.

2. **High-Yield Comparison & Edge Cases**:

- Distinguish between equilibrium/steady-state behavior and transient/limiting behavior.
- Verify dimensional consistency and limiting cases (e.g., as distance or concentration approaches zero or infinity).

3. **Worked Exam Strategy**:

- Write out the known quantities with units, select the exact relation connecting them, solve symbolically first, and then substitute numerical values.